

INTRODUCTION TO POWER BI

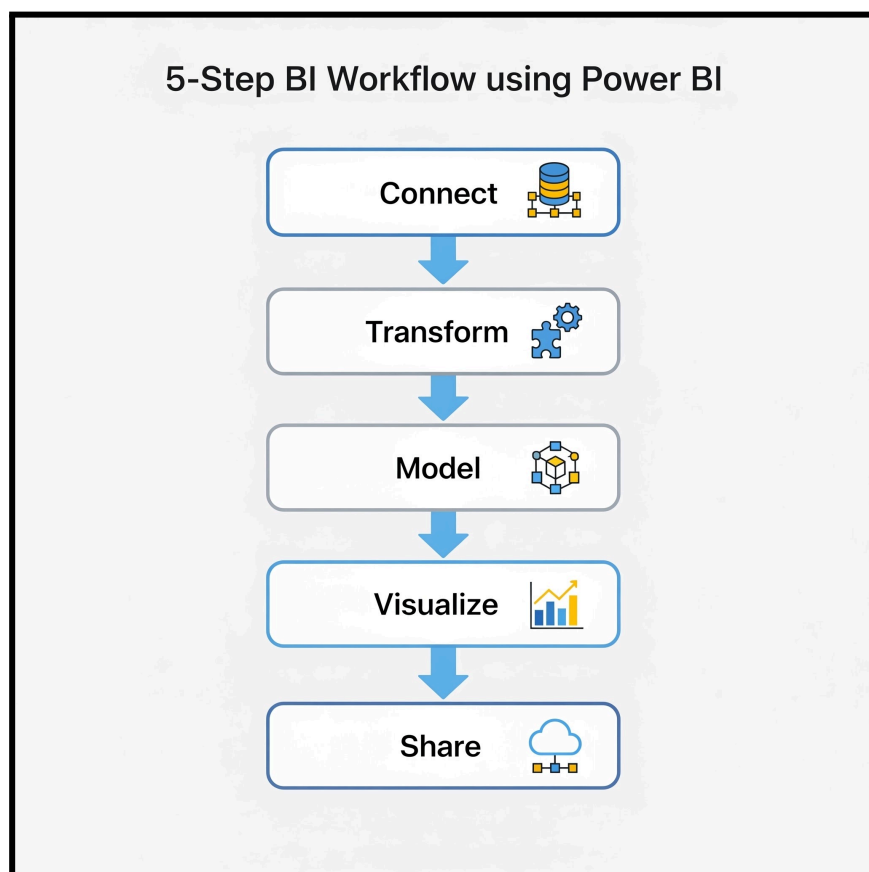
What is Power BI?

- **Definition:** It is a Business Intelligence (BI) tool developed by Microsoft.
- **Purpose:** Its main role is to transform raw, historical data into clear, valuable business insights through data modeling and visualization.

The 5-Step BI Workflow

This is the standard process for extracting insights using Power BI:

1. **Connect -Data Import :** You bring data into Power BI from many different sources (up to 300!), such as Excel files, CSVs, cloud platforms, or local databases.
2. **Transform:** You clean and prepare the raw data using **Power Query**. This process is critical for data accuracy and uses a specific language called M-language.
3. **Model:** You build the data structure by defining and managing relationships between your different data tables. This is done using the formula language called **DAX** (Data Analysis Expressions).
4. **Visualize:** After modeling, you create reports, charts, and dashboards in the Report View to display the findings.
5. **Share:** The visualizations are then shared with stakeholders and businesses via Power BI service.



Understanding the Interface

1. VIEWS

These views serve as your main workspaces within the application

View	Purpose
 Report View	This is the primary area for designing and building all visualizations, charts, and final dashboards.
 Table View	Allows you to inspect the raw data row by row, similar to a spreadsheet, which is useful for checking the data’s structure.
 Model View	This is where you visually manage and define the relationships and connections (Data Modelling) between the tables.
 DAX Query View	A dedicated space for writing and running DAX queries in a tabular format.
 TDML View	A code-based view of the data model that allows for defining model structures and editing model data.

2. RIBBON TABS

The ribbon organizes the tools and features you need for report creation

Tab	Core Function
 File	Manages standard document actions: Open, save, share, import, export, and publishing of Power BI files (.pbix).
 Home	The central hub for frequent actions like importing, performing data transformation, and refreshing datasets.
 Insert	Used to add new elements to your report, such as AI Visuals, Text Boxes, Buttons, Shapes, Images, and Sparklines.
 Σ Modeling	Controls the core data model, including calculation functions (New Measure, New Column, New Table using DAX) and security management (Manage Roles).
 View	Determines the look and appearance of the report, managing themes, page layouts, and toggling the visibility of the Data Pane, Visualization Pane, and Format Pane.
 Optimize	Contains tools specifically designed to improve report speed and performance, such as the Performance Analyzer.
 Help	Provides direct access to documentation, tutorials, and community support.



Installation & Setup Guide

Step 1 — Download Power BI Desktop (Free)

- **Official Link:** Navigate to <https://powerbi.microsoft.com/desktop>.
- **Action:** Select the "**Download free**" option.
- **Compatibility:** This is a native Windows application. Mac users will require a virtual machine (VM) or must utilize the online Power BI Service.

Step 2 — Application Installation

- **Execution:** Launch the installer and follow the standard Windows setup prompts.
- **Launch:** Open the Power BI Desktop application once the installation concludes.